

Embedded Linux & System Software Interview Handbook

Phase 4 – Linux Networking

Linux networking is a fundamental area for kernel developers, cloud engineers, networking engineers and embedded Linux developers. This phase explains how packets travel from user space to the network hardware and back.

Chapter 11 – Linux Networking

Topics Covered

- OSI vs TCP/IP Models
- Linux Networking Architecture
- Socket API
- Socket Types (TCP, UDP, RAW, UNIX)
- System Calls (socket, bind, listen, accept, connect)
- TCP Three-Way Handshake
- TCP State Machine
- UDP Internals
- IP Routing
- ARP
- ICMP
- IPv4 vs IPv6
- Network Namespaces
- Virtual Ethernet (veth)
- Bridge and Linux Bridge
- VLAN
- Bonding and Teaming
- Linux Kernel Network Stack
- Socket Buffers (sk_buff)
- NAPI
- Netfilter
- iptables
- nftables
- Traffic Control (tc)
- eBPF Basics
- XDP (Express Data Path)
- DPDK Overview

- Network Device Drivers
- NIC Interrupt Handling
- RSS and RPS
- Packet Capture (tcpdump, Wireshark)
- Network Performance Tuning
- Debugging Networking Issues
- Real Driver Examples
- Senior Interview Questions

Learning Outcomes

After completing this phase you will understand how Linux implements the TCP/IP stack, how packets traverse the kernel networking stack, how network device drivers interact with the kernel, and how to debug and optimize networking performance in production systems.

Phase	Coverage
Phase 1	Core Linux Kernel Fundamentals
Phase 2	Memory & Process Management
Phase 3	Filesystems & Storage
Phase 4	Linux Networking
Estimated Size	250–350 pages when fully expanded
Target	Senior Embedded Linux / Kernel / Cloud Interviews